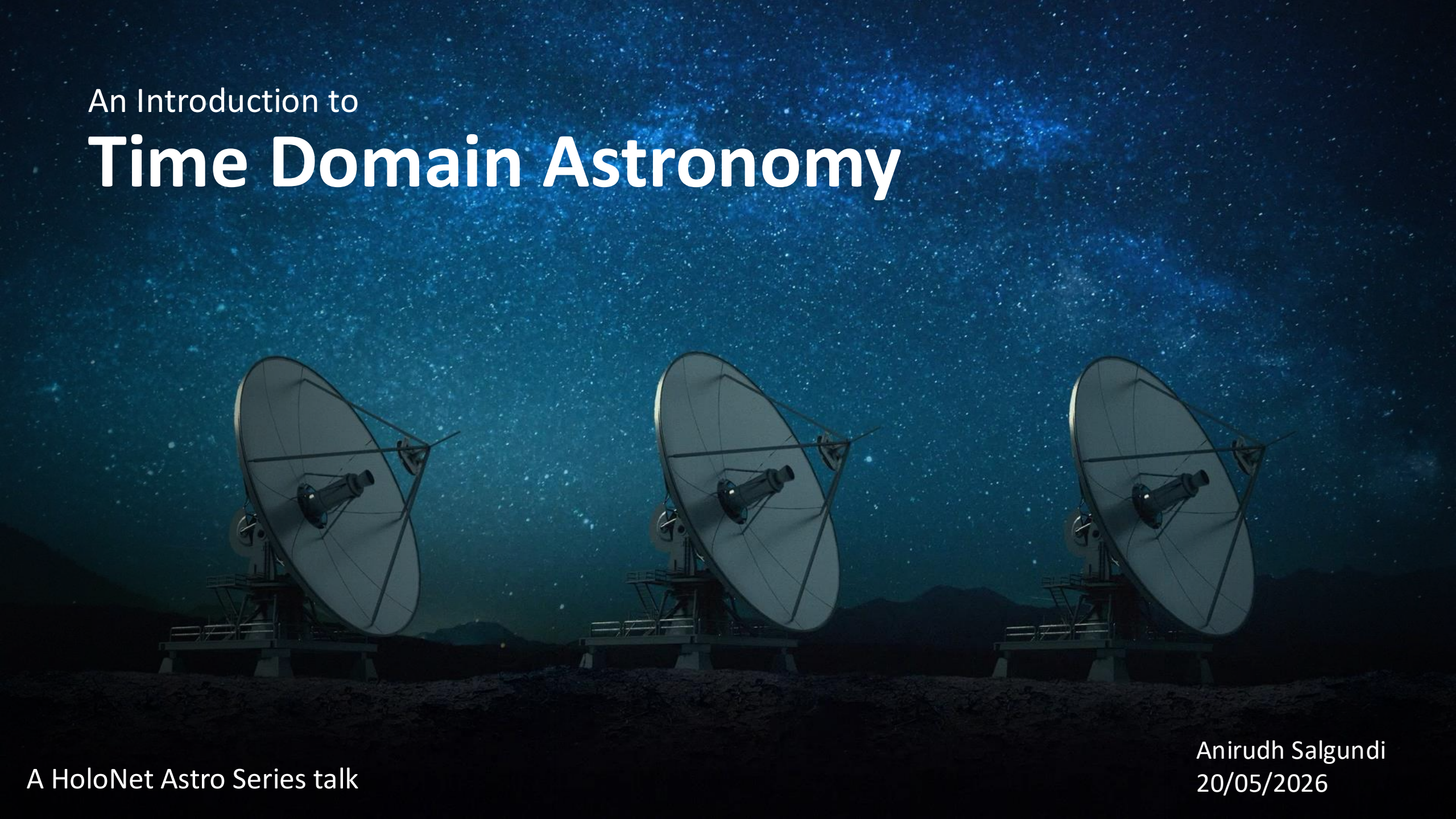
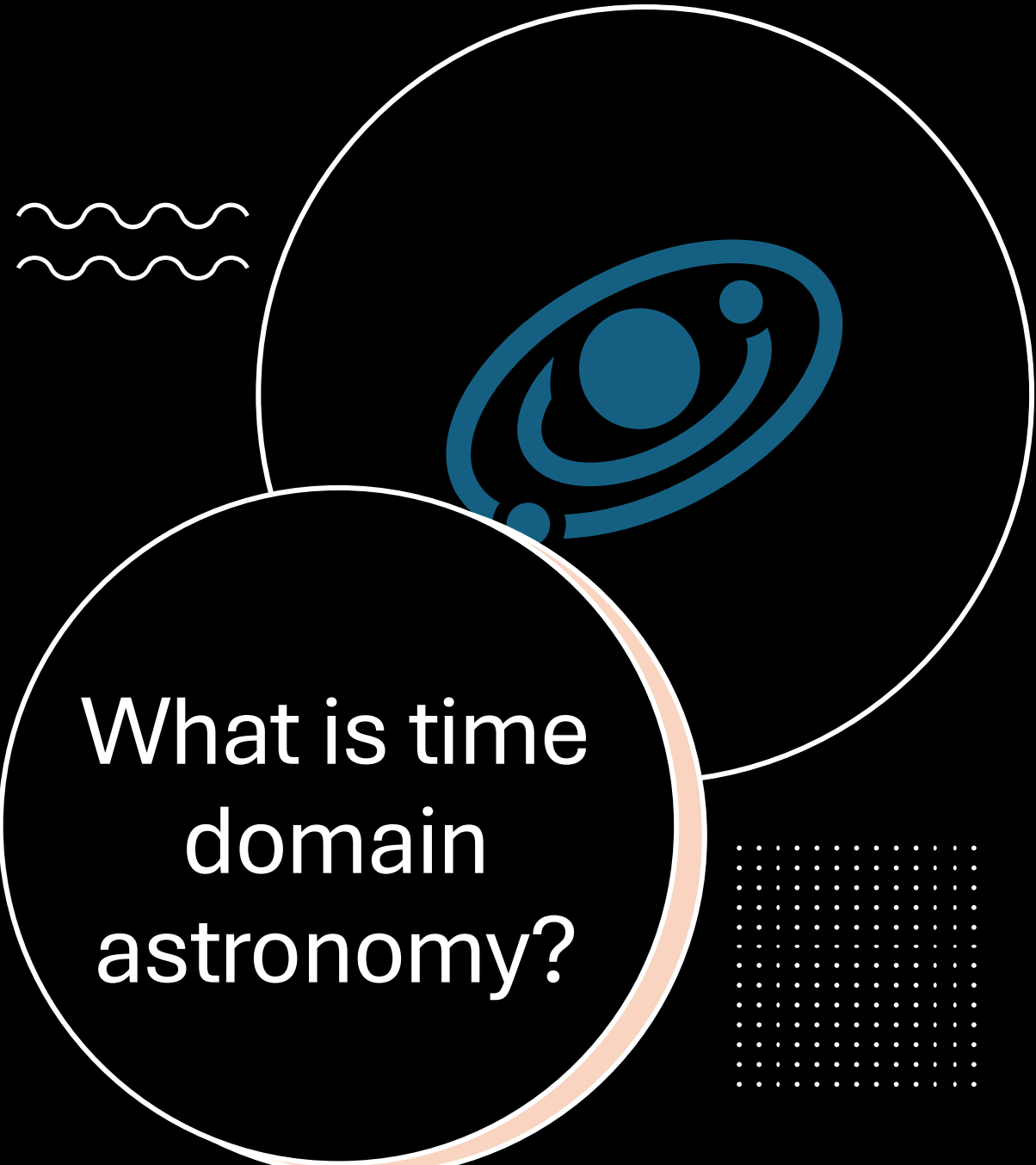


The background of the slide is a photograph of three large radio telescope dishes, likely part of the Arecibo Observatory, pointing towards the sky. The dishes are dark and metallic, with complex support structures. They are set against a deep blue night sky filled with numerous stars and the faint, glowing band of the Milky Way galaxy. The foreground shows a dark, rocky landscape.

An Introduction to **Time Domain Astronomy**

A HoloNet Astro Series talk

Anirudh Salgundi
20/05/2026



What is time domain astronomy?

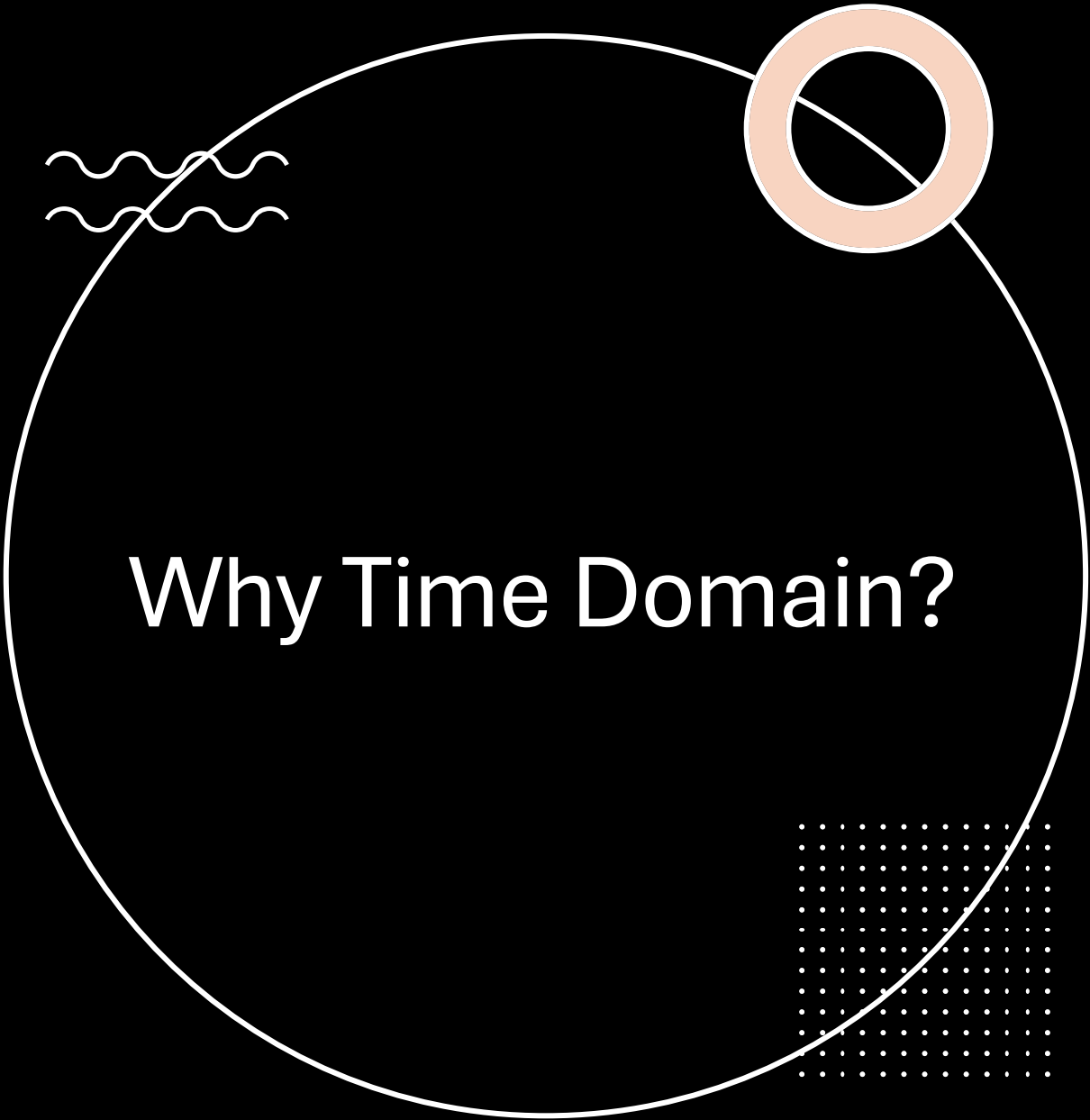
- **Time-domain astronomy is the study of how astronomical objects change with time.**
- What is changing?

Brightness
(Photometry)

Spectrum
(Spectroscopy)

Position
(Astrometry)

Polarization
(Polarimetry)



Why Time Domain?

- Time variability gives access to spatial scales we cannot directly resolve.
- If a source changes significantly over a time Δt , causality requires that information must propagate across the source within that time.

Therefore,

$$R \lesssim c\Delta t$$

Let's do a quick problem

If a source brightens dramatically in 1 second, what is the largest size the source can possibly be?

Do you know of anything at this length scale?

If a source brightens dramatically in:

$$\Delta t \sim 1 \text{ s}$$

then the emitting region must be smaller than:

$$R \lesssim 3 \times 10^8 \text{ m}$$

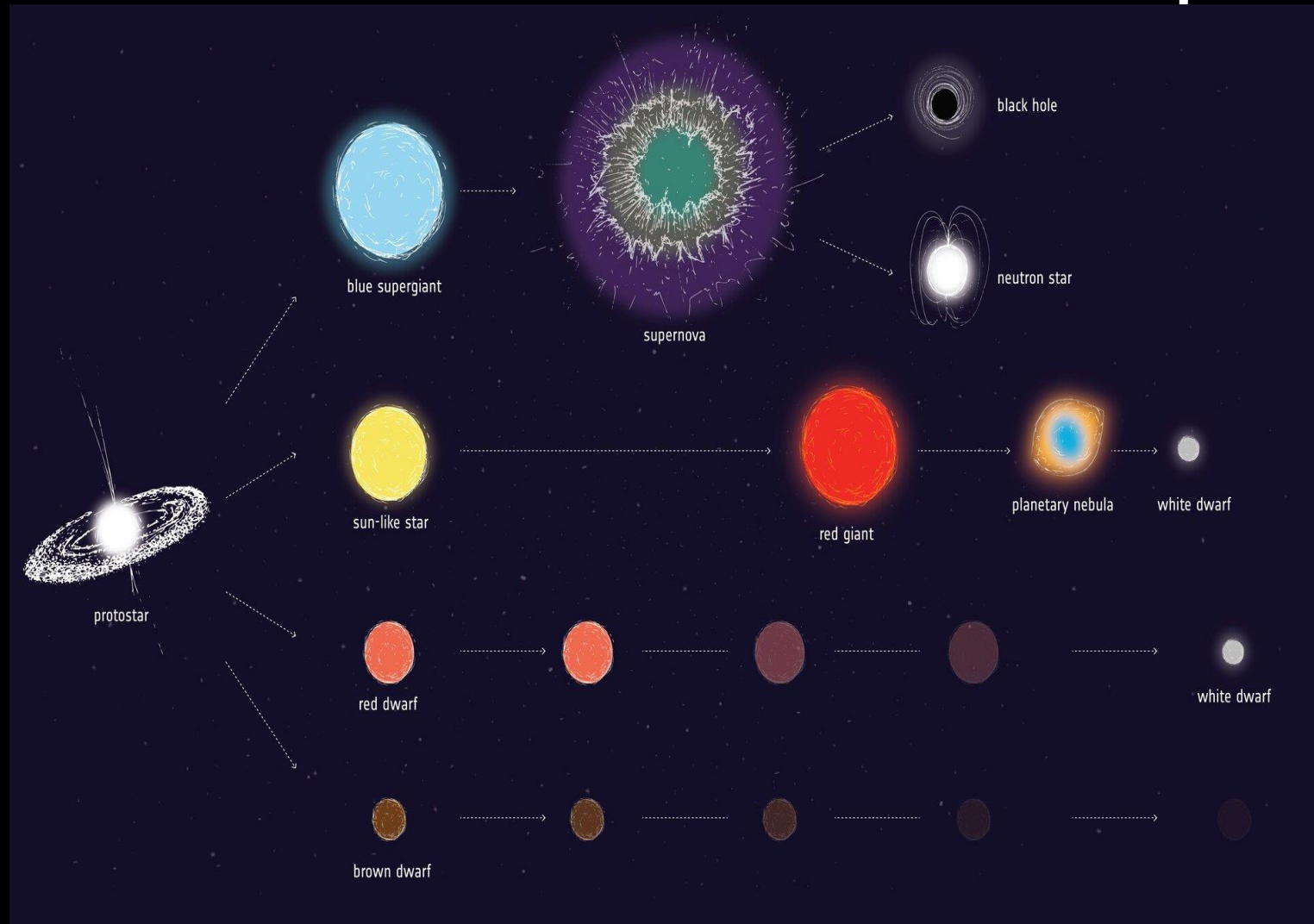
approximately the Earth–Moon distance.

Rapid variability implies compact objects !!

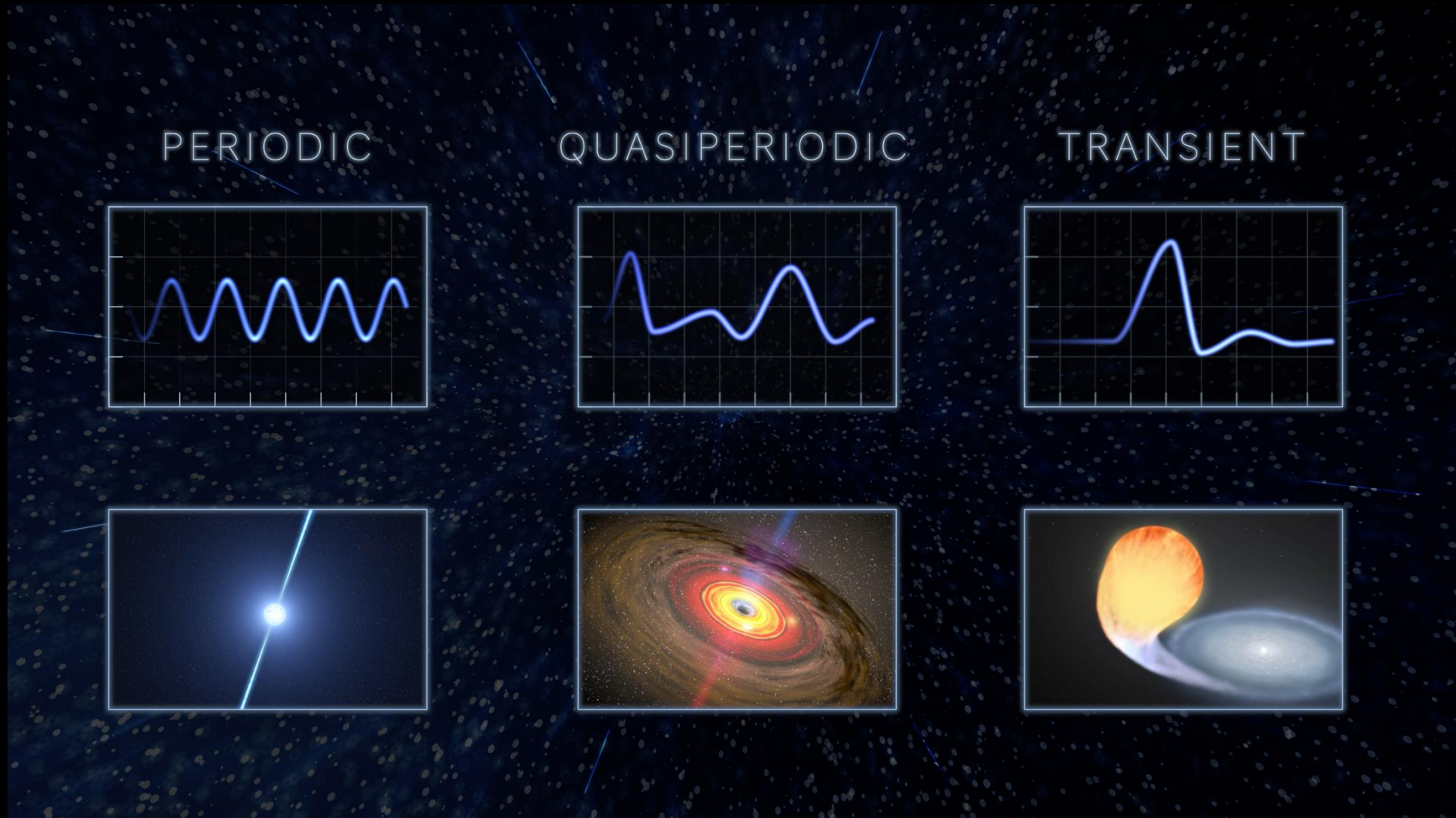
How do we get compact objects?

Quantum Mechanics!

- Electron degeneracy pressure supports white dwarfs
- Neutron degeneracy pressure supports neutron stars
- No stable pressure support remains → black holes



The Time-Domain Sky



Credit: NASA

Observing the dynamic universe

EM SPECTRUM



PULSARS:
Highly magnetized rotating
neutron stars with emission beams



SUPERNOVA AFTERGLOWS:
Expanding stellar ejecta &
transient optical emission



ACCRETION DISKS:
Hot matter orbiting black holes
& neutron stars (X-ray binaries)



GAMMA-RAY BURSTS (GRBs):
Highly relativistic jets from
core collapses

TIME DOMAIN VARIABILITY TIMESCALES

Milliseconds → Seconds → Hours → Days → Years



MILLISECONDS
Fast Radio Bursts (FRBs)



SECONDS
Pulsars



HOURS - DAYS
X-ray Binary Outbursts



DAYS - MONTHS - YEARS
Supernova / Afterglows

VARIABILITY TIMESCALE TRACES
PHYSICAL SIZE: $R \leq c\Delta t$

MULTIMESSENGER ASTRONOMY



PHOTONS
EM TRANSIENT SOURCES:
Probe different wavelengths



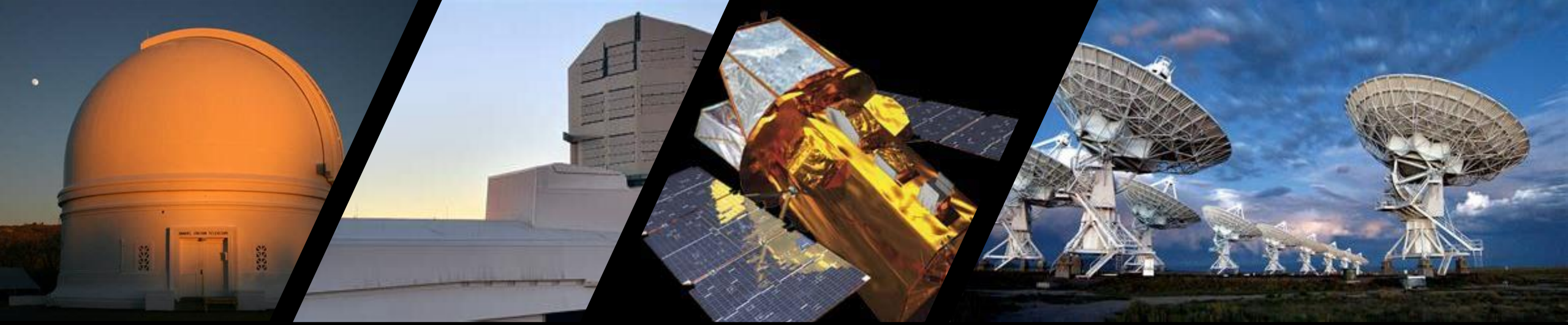
GRAVITATIONAL WAVES
COMPACT BINARY MERGERS:
Neutron star / black hole mergers



NEUTRINOS
CORE-COLLAPSE SUPERNOVAE:
High-energy astrophysical sources

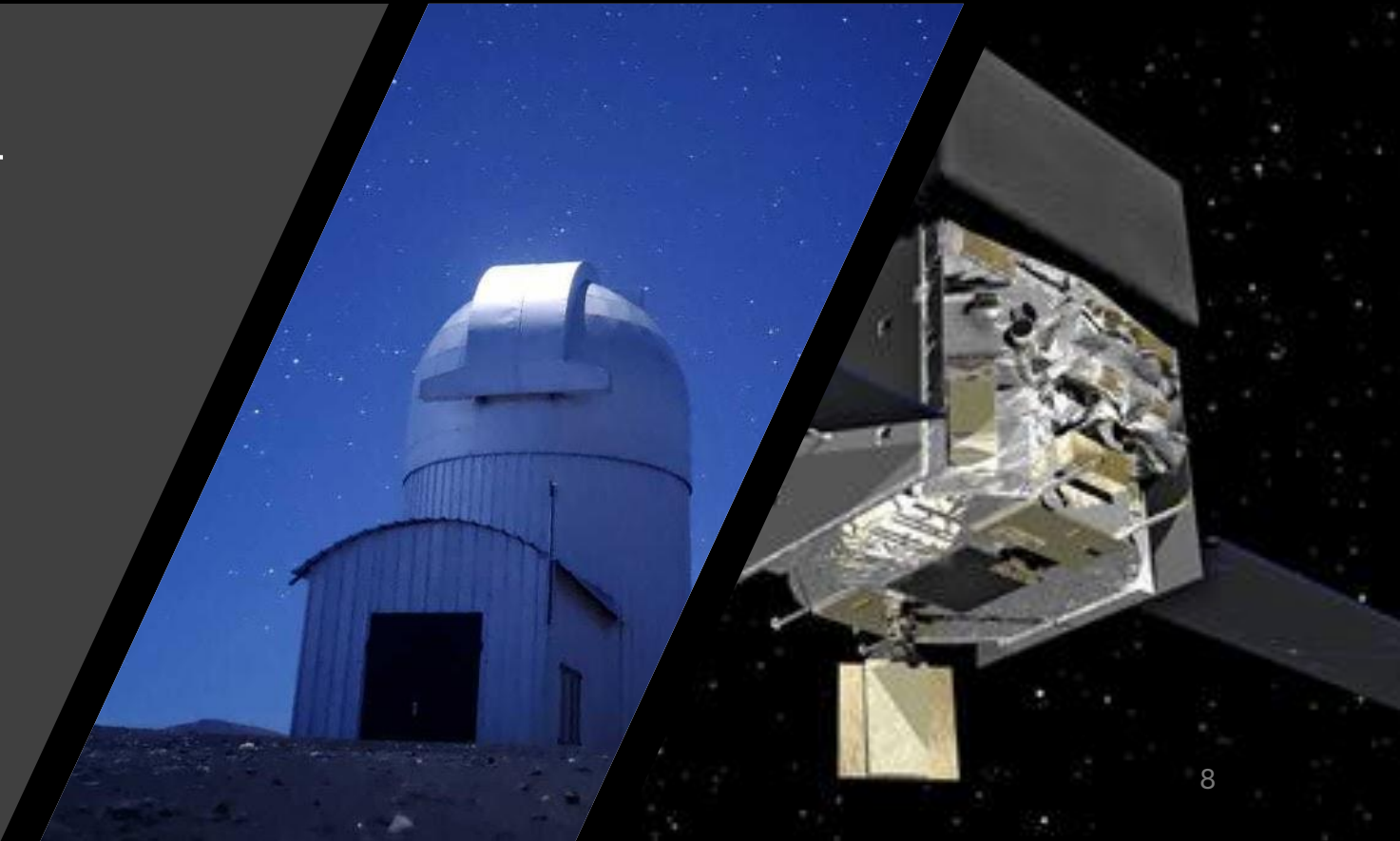


Image Credit: Gemini AI



How do we get these astronomical data?

Big survey telescopes and small robotic telescopes!



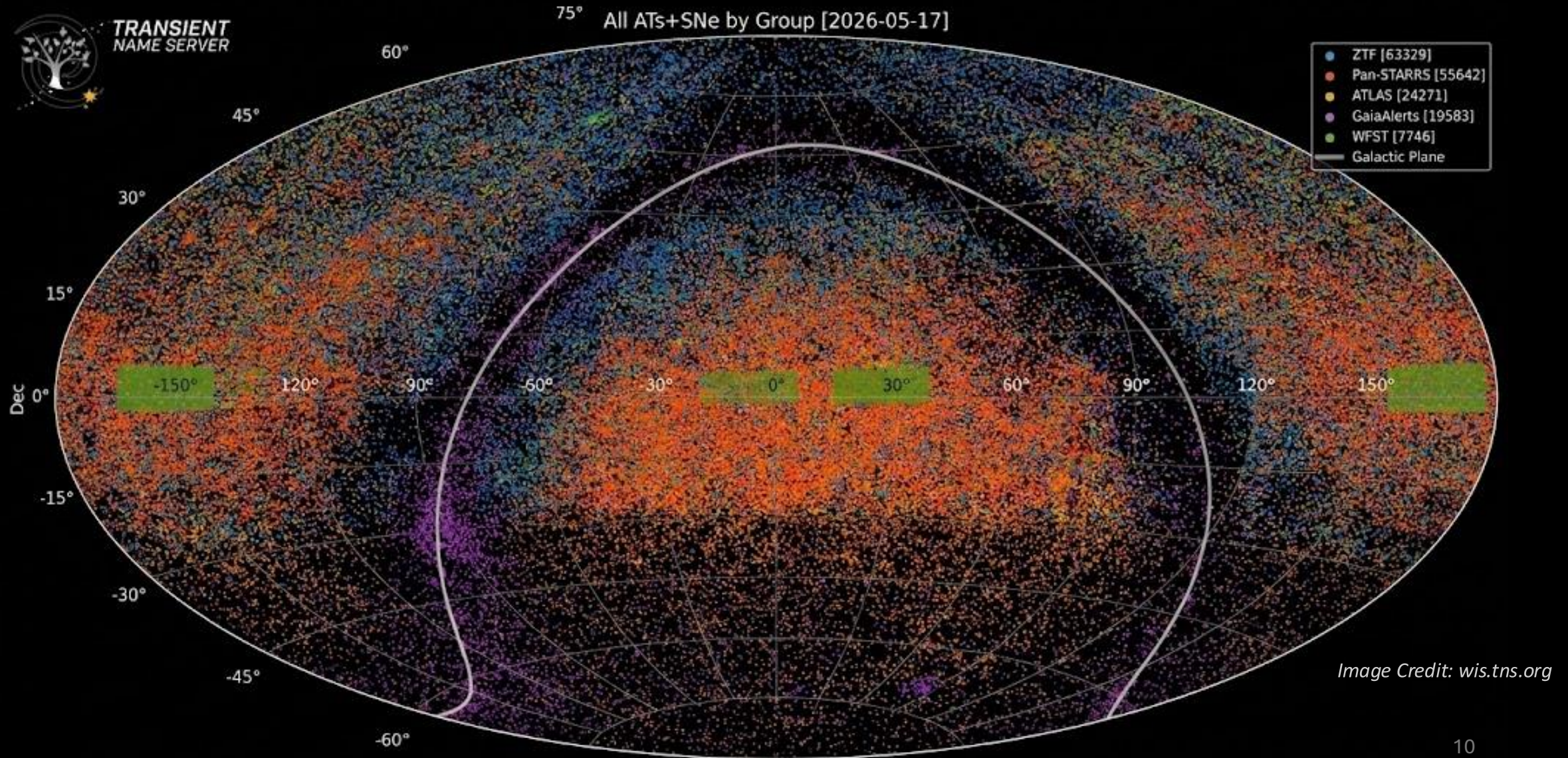


Global Relay of Observatories Watching Transients Happen

Growth India @IITB!

Image Credit:
growth.caltech.edu/

The transient sky as of 17th May 2026 !!



The Zoo of Compact Object Transients

Categorizing by the fundamental physical mechanisms driving their extreme energy output

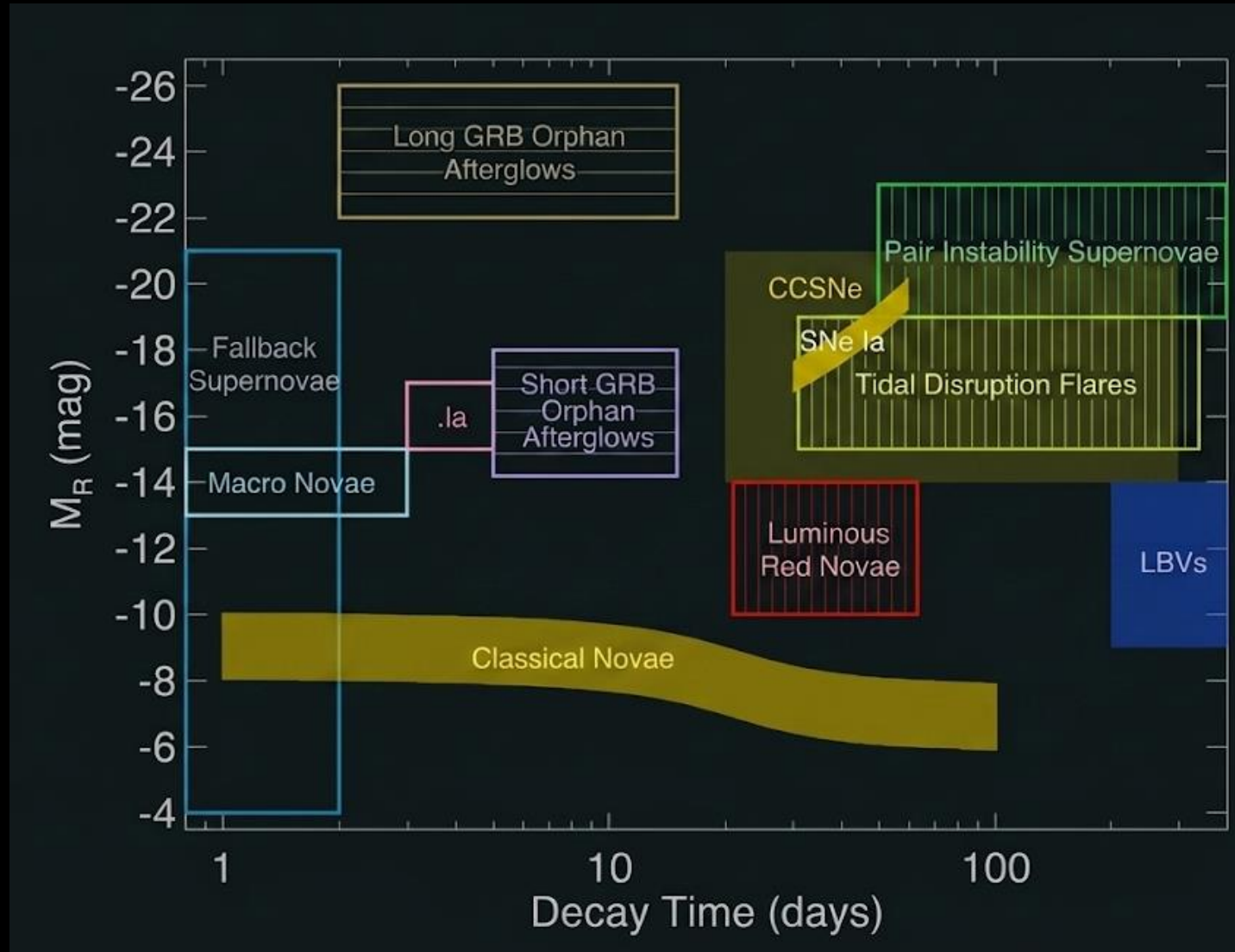


Image Credit: mdpi.com



Mergers of compact objects (NS-NS, NS-BH)

Confirms r-process nucleosynthesis
(heavy element creation) and
constrains the ultra-dense neutron
star Equation of State (EOS).

Mergers and tidal disruptions.

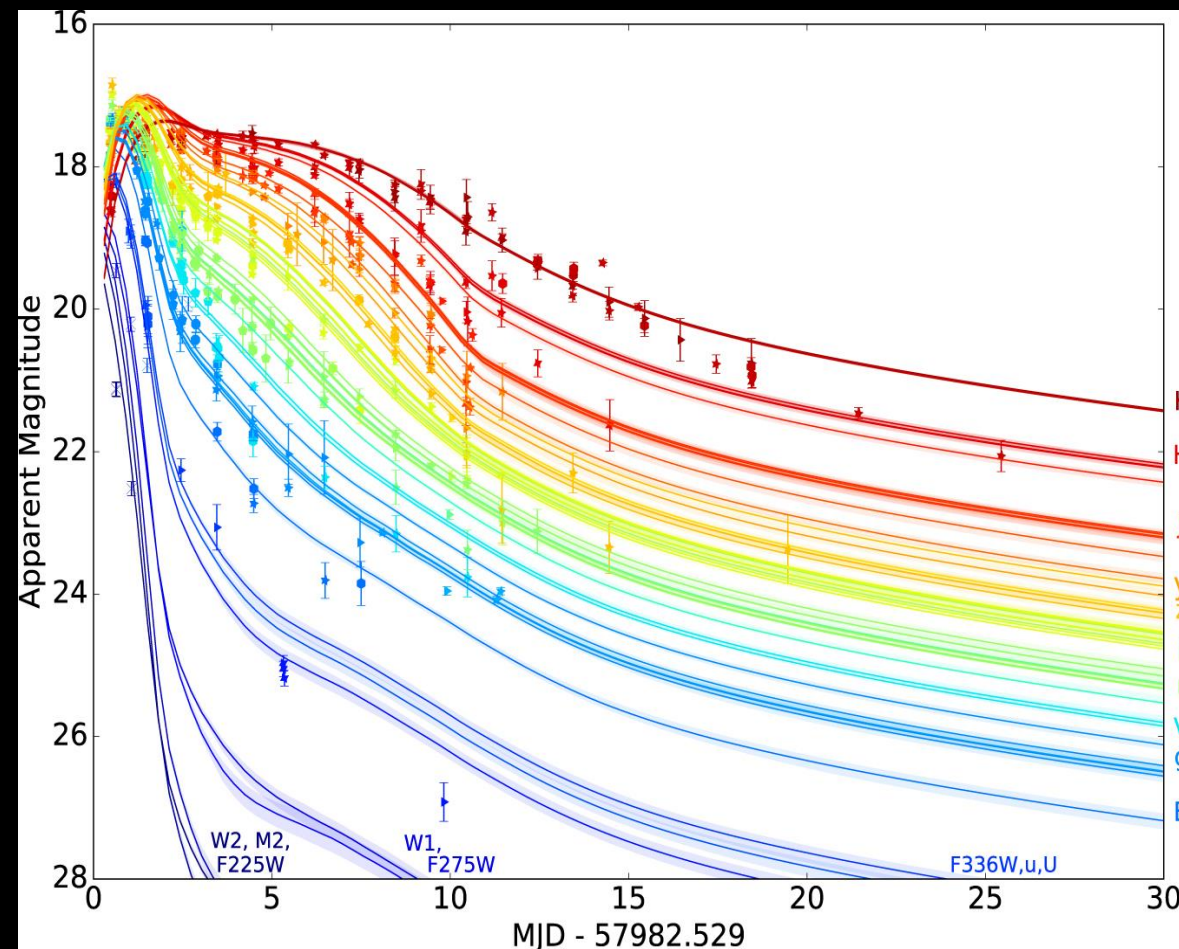
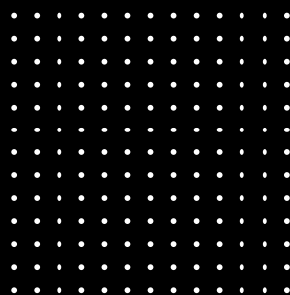
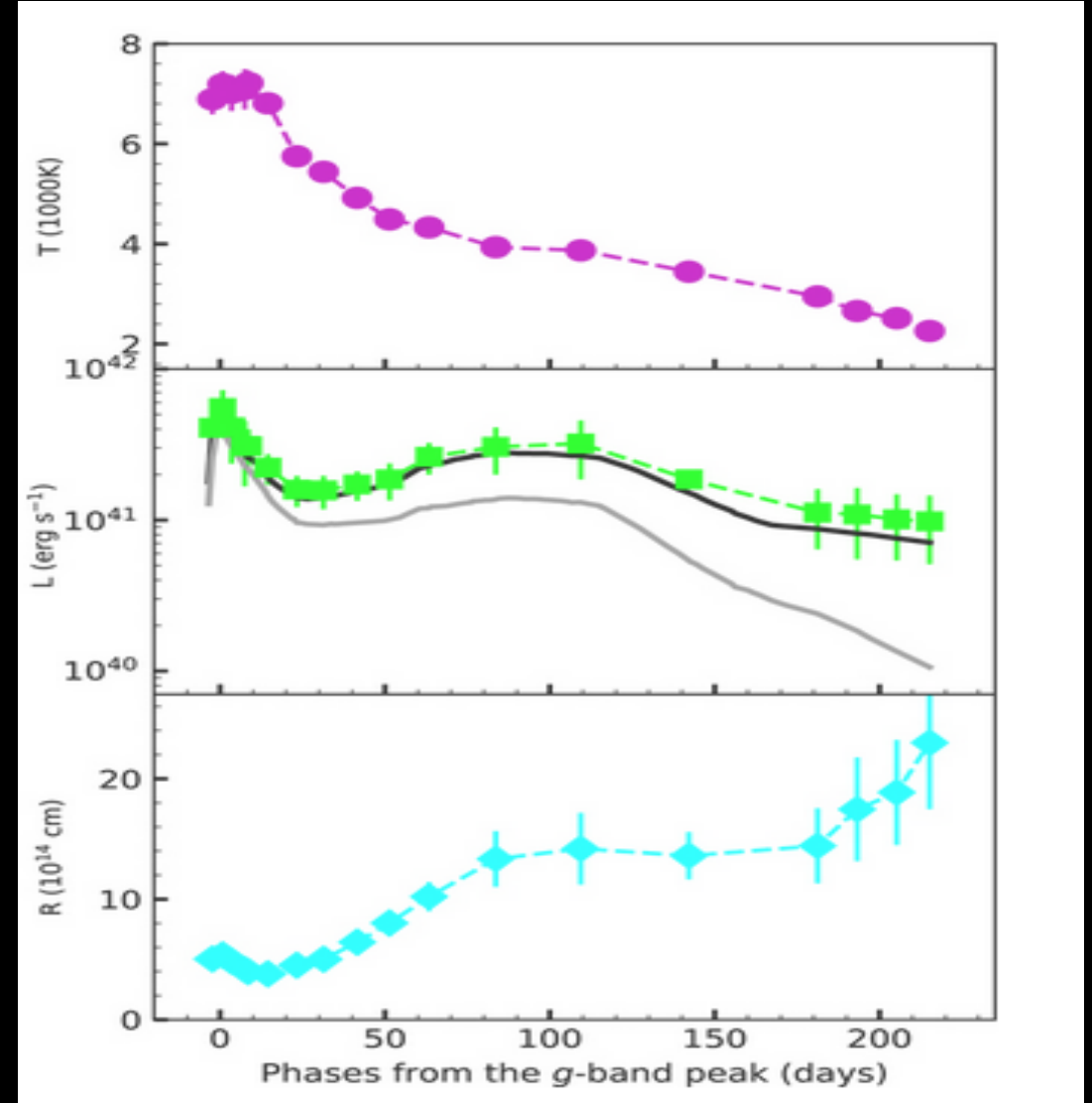


Image Credit: Villar, ., +2017

Stellar Mergers

Provides a real-time laboratory for stellar envelope hydrodynamics and the highly uncertain physics of binary star fusion.

Mergers and
tidal
disruptions.





Tidal Disruption events

Maps the mass and spin of dormant supermassive black holes and traces the eccentric fallback dynamics

Also helps us in understanding what can drive relativistic jets

Mergers and tidal disruptions.

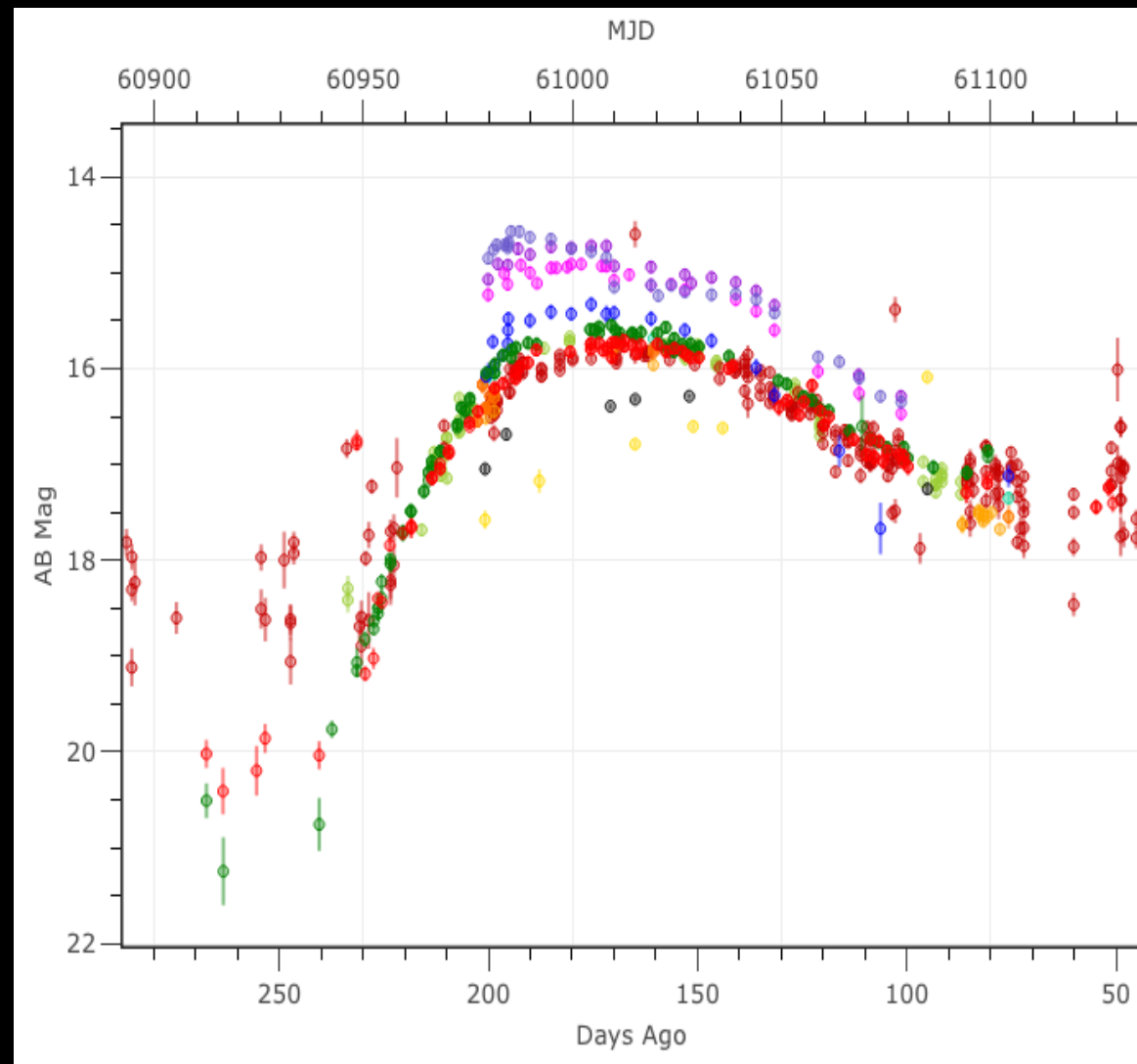
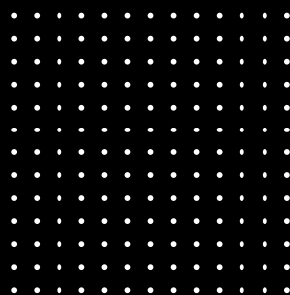


Image Credit: fritz.science

Gamma Ray Bursts

Tests Lorentz invariance across cosmic distances and maps the conditions required for a collapsing star to sustain a relativistic jet.

Relativistic
jets and
shocks!

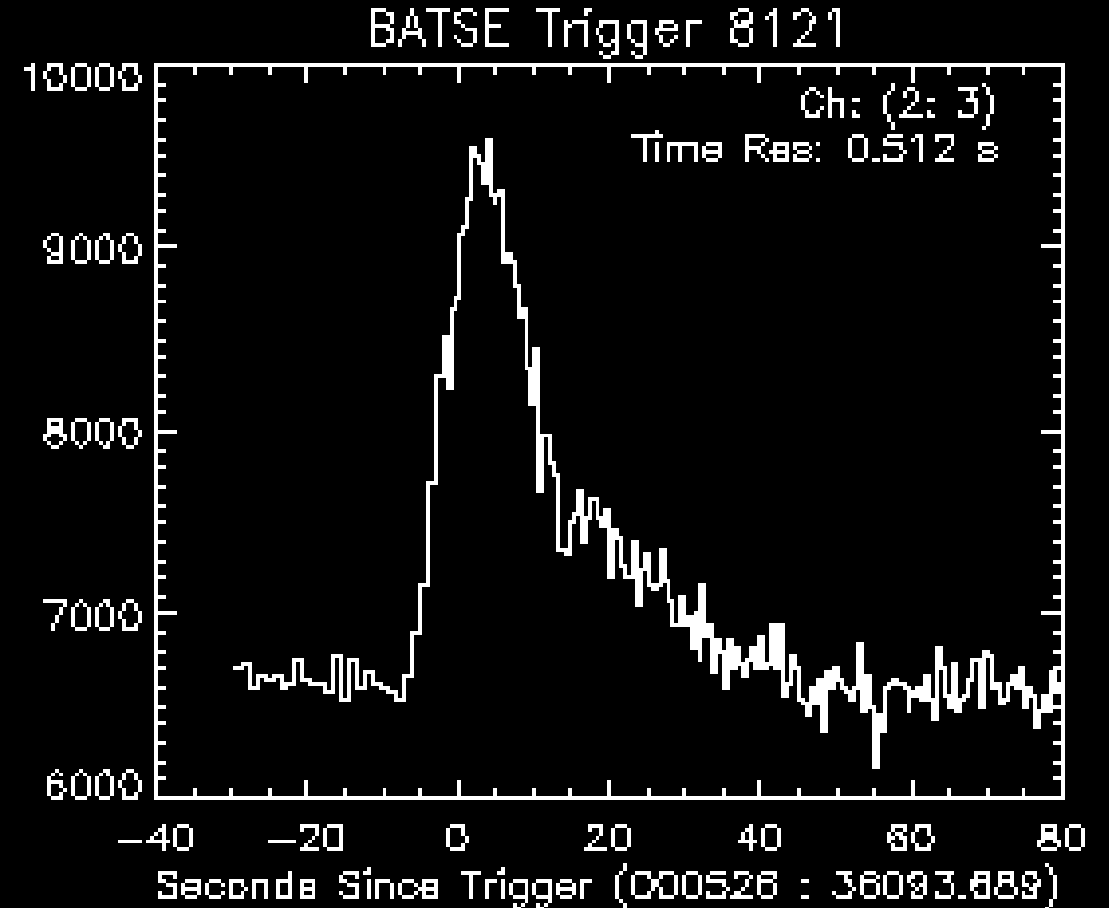


Image Credit: BATSE Team

Supernova

Constrain the progenitor star's
radius and do cosmology!

Relativistic
jets and
shocks!

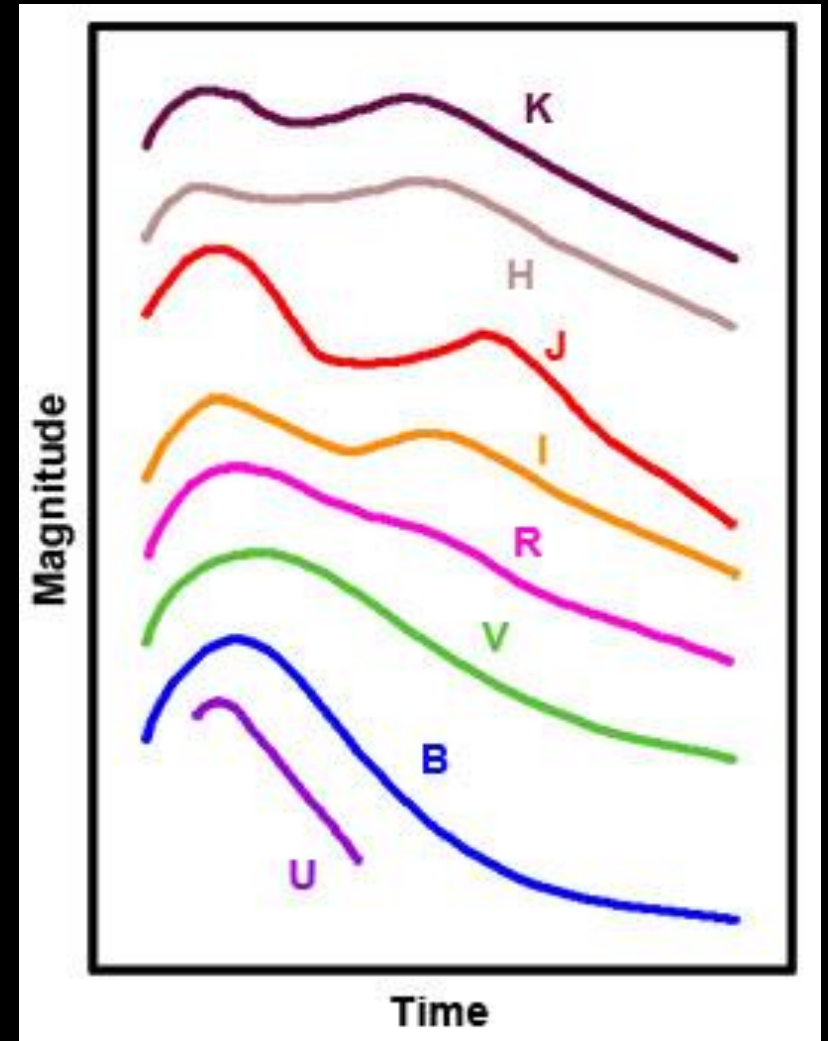
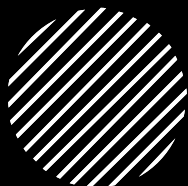


Image Credit: Google

Accretion driven and thermonuclear instabilities



Cataclysmic variables and Novae

Tests binary evolution models,
maps explosive nucleosynthesis
pathways, and defines progenitor
mass limits for Type Ia supernovae.

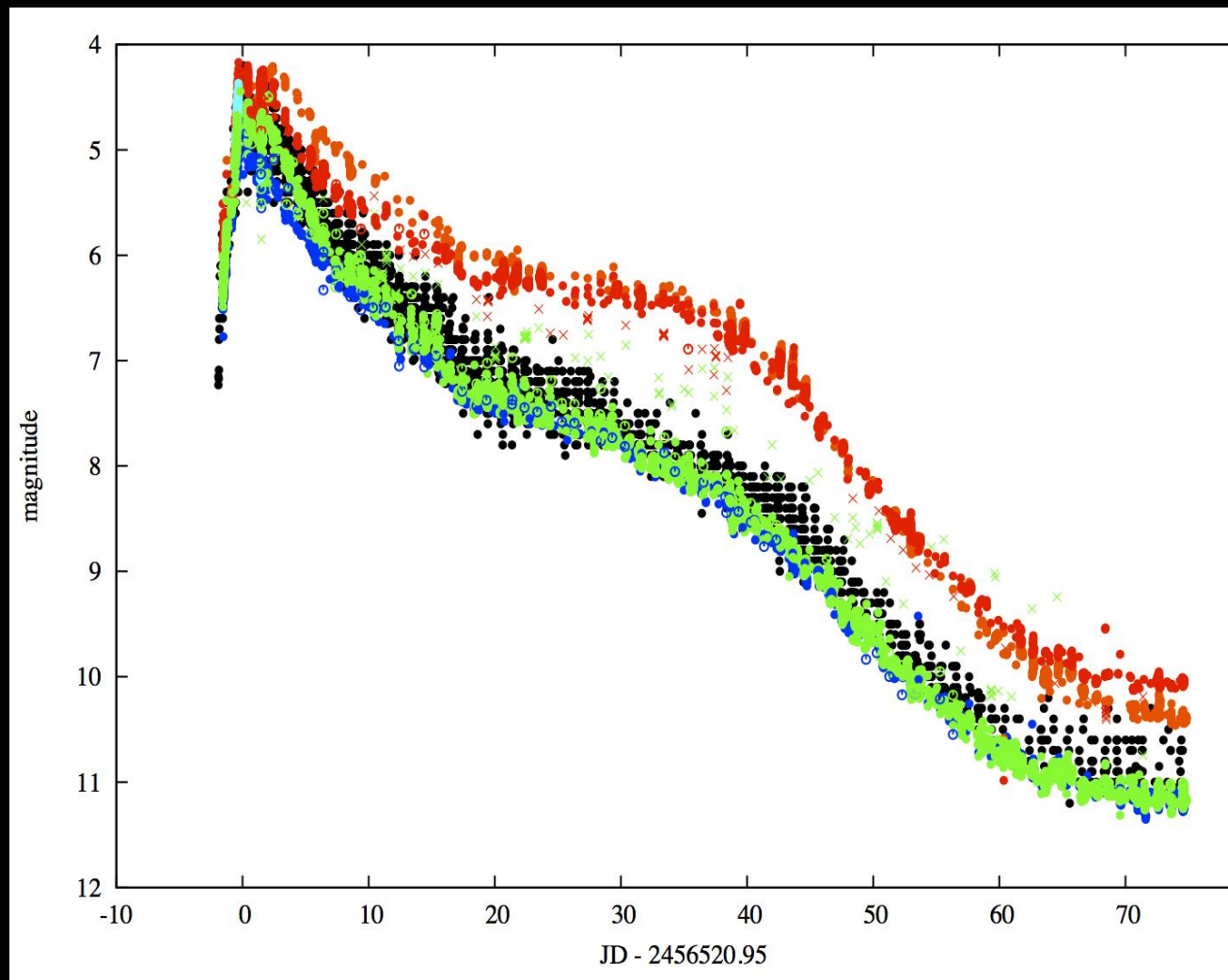


Image Credit: astrobit.es.org

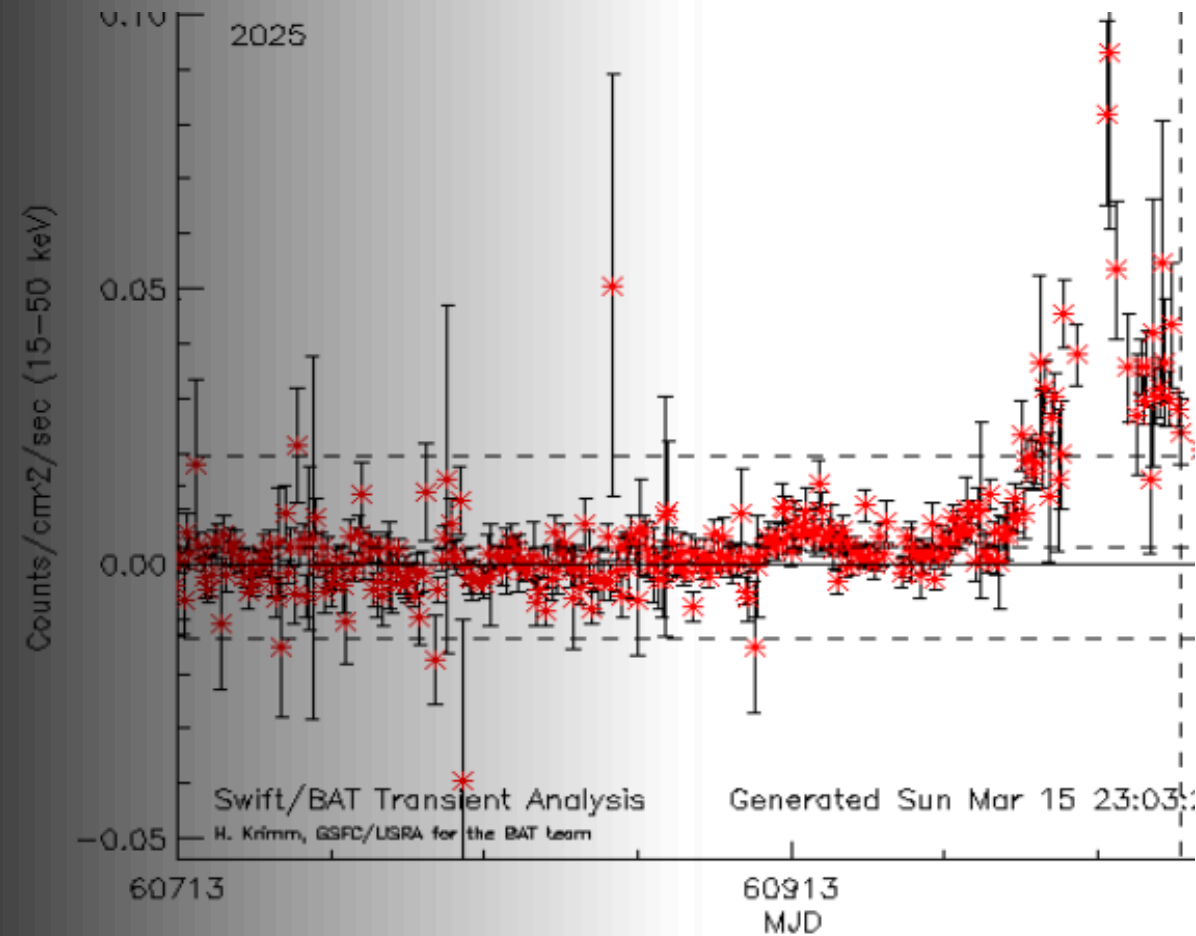
Accretion driven and thermonuclear instabilities

X-ray Binaries

Probes Plasma physics under extreme magnetic fields!

The only class of objects we discussed today that show all of periodic, Quasi Periodic and Transient features!

Image Credit: swift BAT



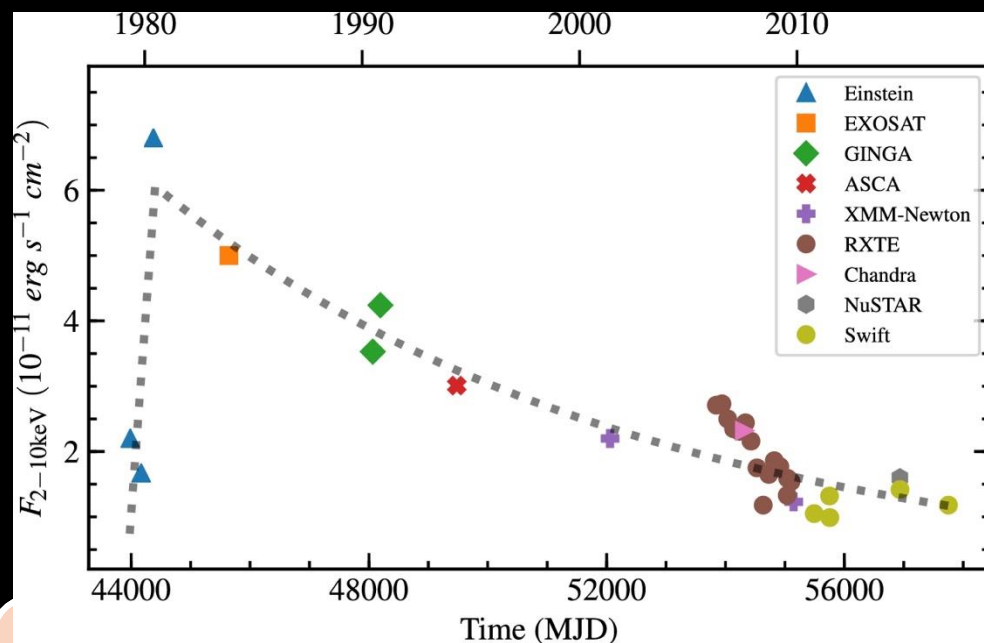


Image Credit: Zhen, Y., 2018

Let's pause for a second

- What did all the lightcurves have in common?
- A Fast Rise and an Exponential Decay! (FRED)

Typically:

- **fast rise** → sudden energy injection
- **slow decay** → cooling, diffusion, expansion, or depletion

Communication
is the key!

The Astronomer's Telegram

Post | Search | Policies
Credential | Feeds | Email

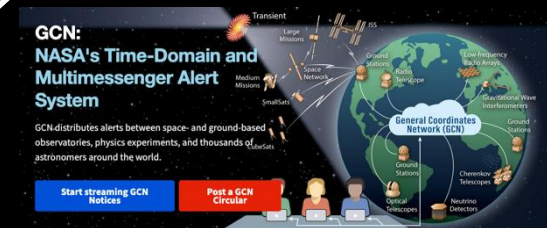
20 May 2026, 02:27 UT

 **Enable Notifications**
You have no devices connected

X-ray Binaries, AGN,
Novae, Flaring Stars



191,985
Supernovae and
Optical Transients!



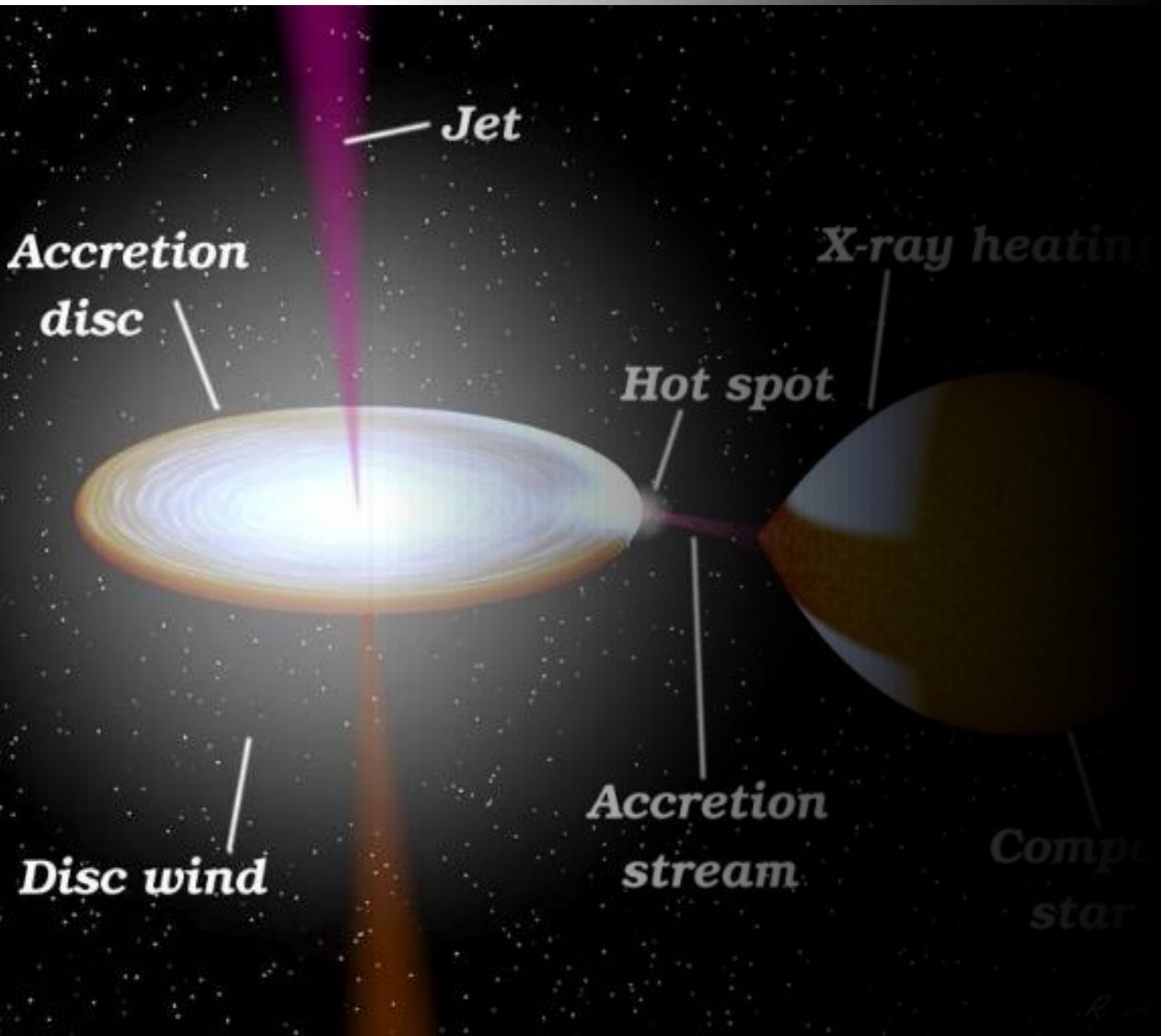
GRBs, Gravitational
Waves, Neutrinos





Archival data treasure

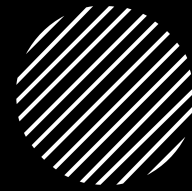
Not all data have been fully utilized!
Literally goldmine of data!



X-ray binaries

- Compact object (NS/BH) and a companion star.
- Highly luminous in X-rays, which are produced by the accretion of stellar material from the companion star.

Image Credit: cosmos.esa.int



Soft X-ray Transients

- Can undergo outbursts (days to months!)
- Can undergo thermonuclear bursts (Seconds to Minutes!)

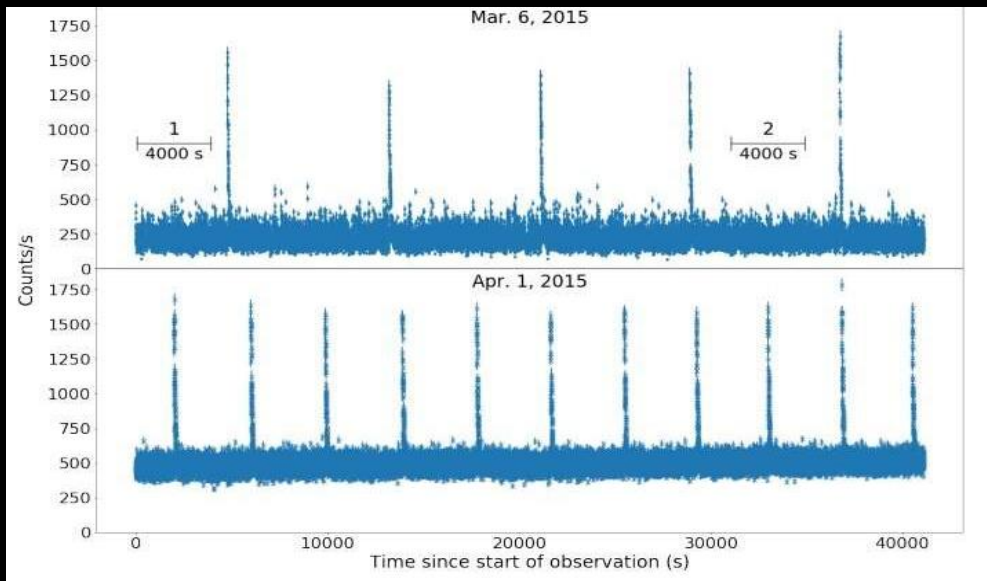
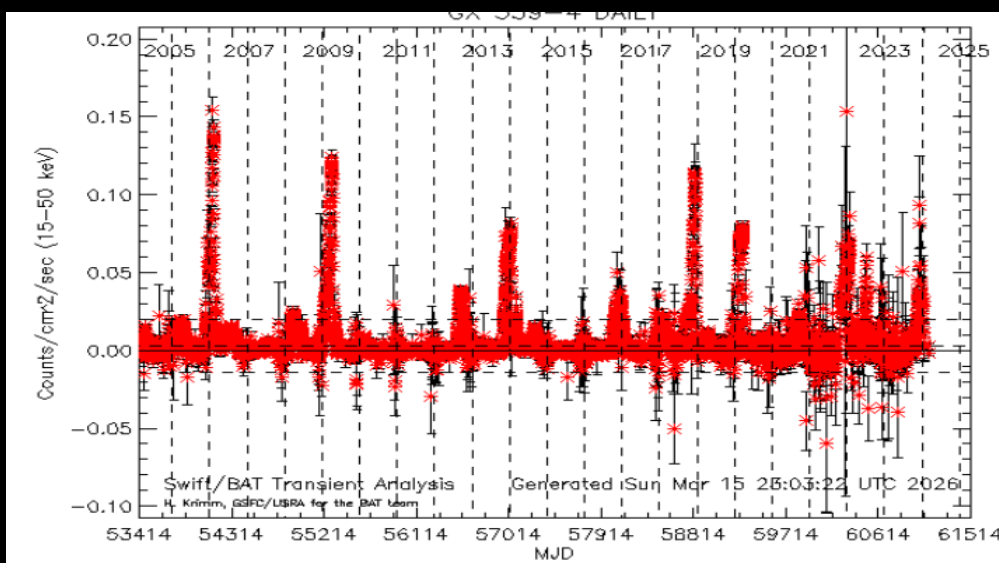
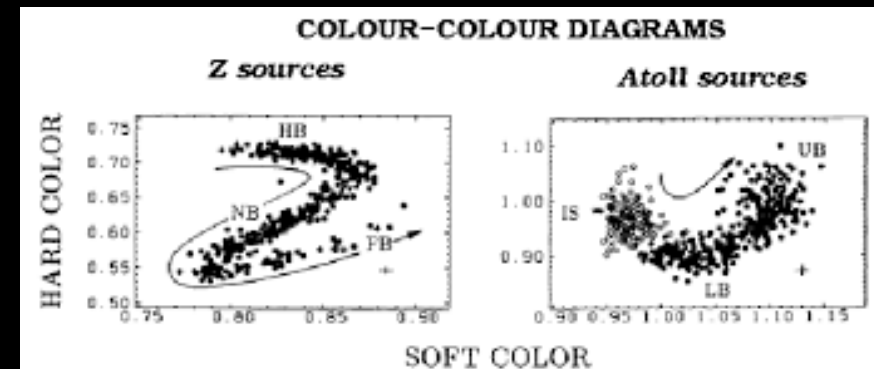
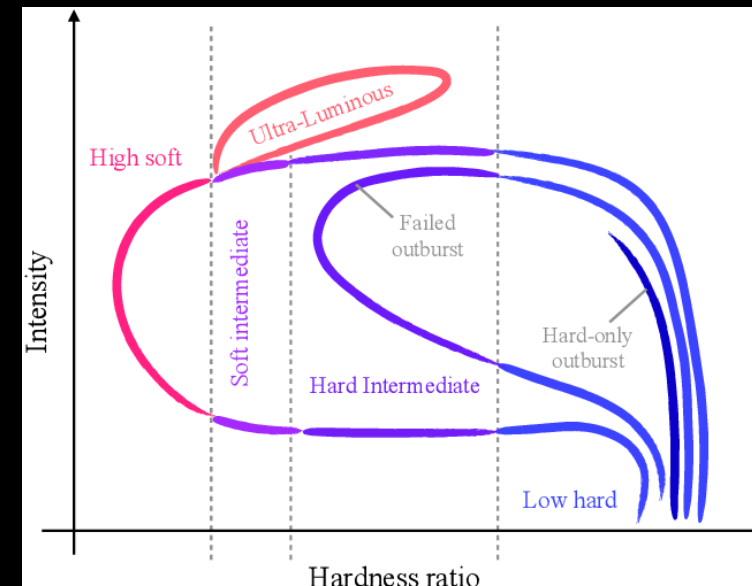


Image Credit: Swift BAT

But not all outbursts are same!

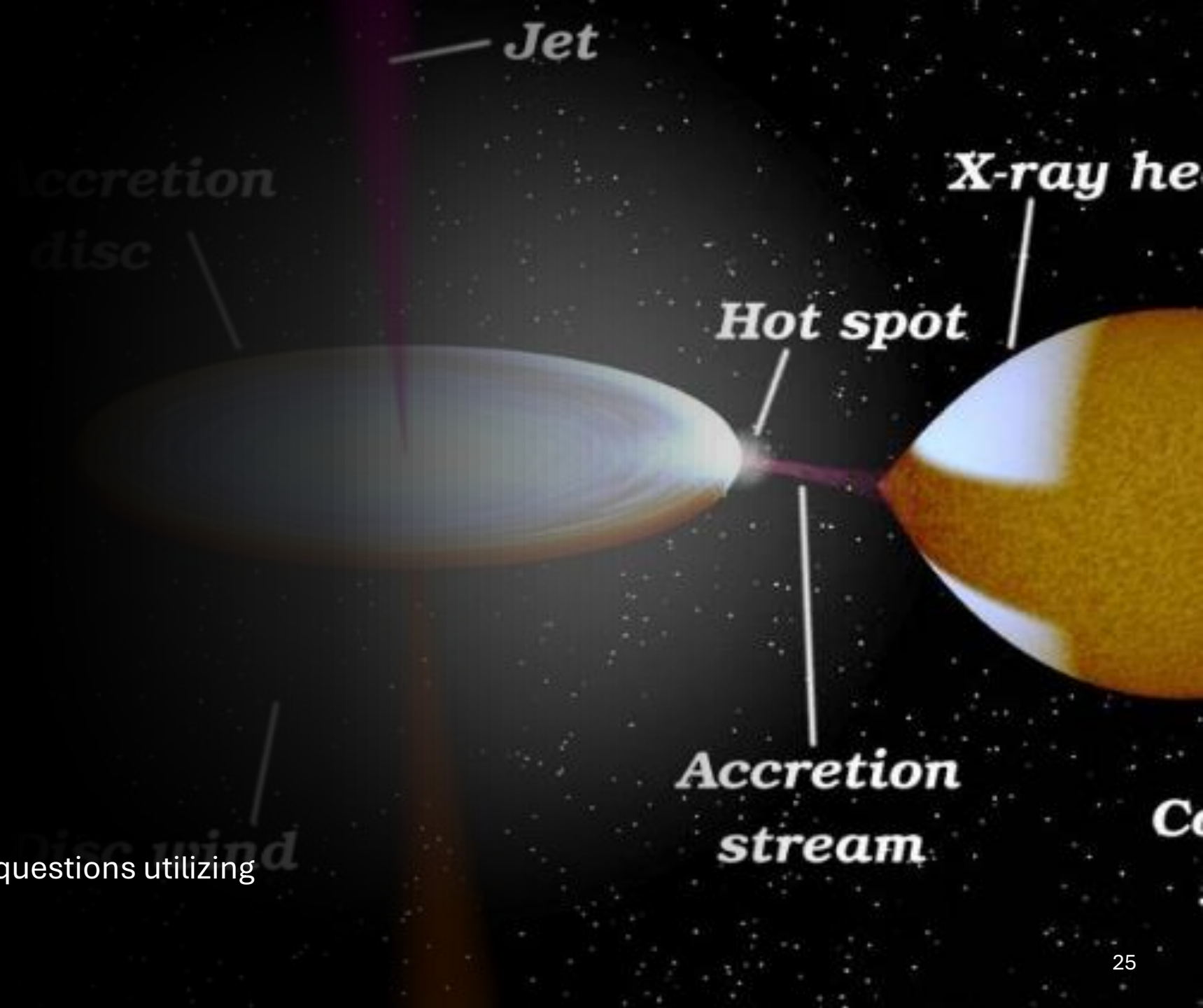
Black Holes and Neutron star X-ray Binaries behave very differently in an outburst!



But physical Scenario is the same

- What is different?
- What drives these differences?
- Do the lightcurve look similar in all wavelengths?
- Are all outbursts of these sub classes same?

We will try to answer some of these questions utilizing archival data in the KSP project!





Thank you!